3.5 Data processing

Sensors produce raw data streams that need processing, decoding, and protocol transformation before they are transported into your digital twin. We will deal with data ingestion and storage in chapter 4, but prior to ingesting the data, we must decode it from sensor specific formats and protocols keeping in mind the unique characteristics of IoT sensor data.

3.5.1 Understanding IoT data characteristics

There are distinct attributes of the sensor data originating from IoT devices that set it apart from traditional data, impacting how it's handled in digital twin applications. These key characteristics can be summarized in the following way.

VOLUME

IoT devices may generate vast quantities of data, from continuous sensor readings, to extensive log files, requiring scalable storage and processing solutions.

VELOCITY

Data is produced and transmitted at high speeds, often in near real-time. This demands rapid processing to keep digital twins continuously updated and responsive. Often the velocity of data is variable requiring a processing architecture that can scale elastically.

VARIETY

IoT data comes in diverse formats and structures, originating from various sensor types and systems. Every sensor manufacturer has their own data format. Some send human-readable JSON:

```
{"temp": 23.5, "humidity": 45.2, "battery": 3.2}
```

Others pack data into binary formats to save bandwidth: